Durham University

Examination Paper

Examination Session:	Year:	Exam Code:
May/June	2020	ENGI30420-WE01

Title: MSc Advanced Mechanical Engineering/MSc New and Renewable Energy

Thermodynamics & Fluid Mechanics 3

Time Allowed:	2 hours	
Additional Material provided:	Compressible Flow Data Booklet. Data Tables for Thermodynamics.	
Materials Permitted:	None.	
Calculators Permitted:		Models Permitted: Those from the Casio fx-83 and fx-85 series.
Visiting Students may use dictionaries: YES		

Instructions to Candidates:	Answer ALL questions. Use a separate answer book for each question. All relevant workings must be shown.

Revision:	

Q1

A repeating stage axial flow steam turbine is designed to have constant axial flow velocity.

(a) Sketch a cascade view and the velocity triangles. Show that

$$\psi = 2\left(1 - R - \phi \tan \alpha_1\right),$$

where ψ is the stage loading coefficient, R is the reaction, ϕ is the flow coefficient and α_1 is the interstage swirl angle, i.e. the stator inlet flow angle.

[40%]

(b) This turbine is to have an overall enthalpy drop of 200 kJkg^{-1}. It has a mean blade speed 180 ms^{-1}, 35$\%$ reaction, a flow coefficient of 0.4 and 20° interstage swirl. Calculate the number of stages required to achieve this overall enthalpy drop.

[20%]

(c) The total-to-total efficiency of each stage in part (b) can be estimated as

$$\eta_{tt} = 1 - \frac{0.05}{\Delta h_0}\left(V_2^2 + W_3^2\right),$$

where V_2 is the absolute velocity at the stator exit, W_3 is the relative velocity at the rotor exit and Δh_0 is the change in specific total enthalpy across the stage. Calculate the total-to-total efficiency η_{tt} for this turbine stage.

[25%]

(d) Calculate the specific kinetic energy of the flow at the rotor exit and hence calculate the total-to-static efficiency η_{ts}.

[15%]

continued

Q2

(a) Give pros and cons of the nuclear power when compared to the other major kinds of power plants.

[10%]

(b) Using the nuclear data provided in Table Q2, determine the products of radioactive decay for the following nuclei:
i) H-3→
ii) O-15→

[10%]

(c) In a nuclear reactor a neutron with kinetic energy of 200 keV collides with an atom of U-238. If the critical energy for fission in U-239 is 5.5 MeV and using the nuclear data given in Table Q2, determine whether fission can occur. The equivalence between mass and energy on an atomic scale is: 1 u = 931.5 MeV.

[15%]

Element	Symbol	Atomic number (Z)	Atomic mass(u)
Proton	p	1	1.007277
Neutron	n	0	1.008665
Electron	e	0	0.000550
Hydrogen	^{2}H	1	2.014102
Hydrogen	^{3}H	1	3.016049
Helium	^{3}He	2	3.016029
Helium	^{4}He	2	4.002602
Nitrogen	^{14}N	7	14.003074
Nitrogen	^{15}N	7	15.000109
Oxygen	^{15}O	8	15.003065
Oxygen	^{16}O	8	15.994915
Fluorine	^{16}F	9	16.011465
Fluorine	^{17}F	9	17.002095
Uranium	^{238}U	92	238.050788
Uranium	^{239}U	92	239.054294

Table Q2

(d) Describe the main characteristic features of the PWR reactor and compare its size to other technologies.

[15%]

(e) Plasma temperatures of approximately 300 million degrees are required to sustain a fusion reaction. Why is the temperature required to sustain fusion so much higher than for fission and why is the Sun able to sustain reaction at only 15 million degrees?

[15%]

continued

(f) Sketch typical steam expansions through the high-pressure turbines of PWR and of AGR nuclear power cycles on the same T-s diagram. Comment on the steam conditions at the inlet and at the exit from each of these types of high-pressure steam turbine.

[20%]

(g) Compare the typical electric power outputs from PWR and AGR nuclear power cycles. Explain the differences in the steam outputs from the nuclear reactor steam generators in these two types of power cycle, that produce the different power output levels.

[15%]

Q3

A supersonic fighter aircraft is optimised for cruise at a Mach number of 2.0 at an altitude where the ambient pressure is 0.07 bar. During development, the engineers need to take into consideration the weapons carried by this aircraft. In the current design stage, the aircraft is to carry one missile underneath each of the two wings as sketched in Fig. Q3 for one wing. It is reasonable to assume the missile body ABCDEF to be two dimensional and has a width of 80mm (into the page).The missile body is symmetric about the line AD, which is at zero incidence. $\angle FAD = \angle EDA = 20°$, $\angle ABC = 157°$. Other key dimensions are labelled in Fig. Q3. Neglect the effect of the wing and the pylon geometry.

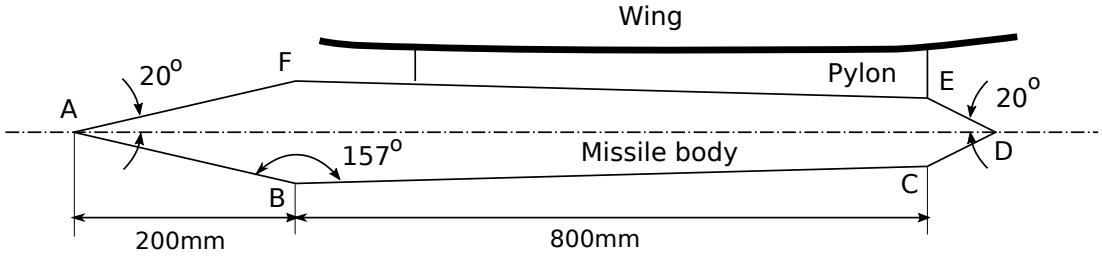


Fig. Q3: Not to scale

(a) Sketch and label the aerodynamic features developed on the missile body. Provide basic justifications without detailed calculation.

[30%]

(b) Calculate the increase in the aircraft engine cruise thrust required to overcome the additional pressure drag of the two missiles.

[70%]

END